De novo short read assembly

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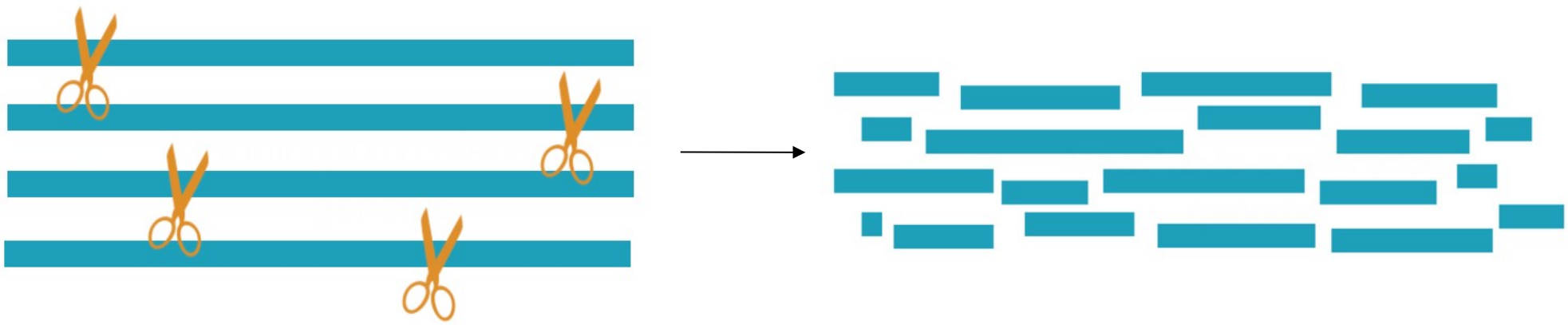
Sequence assembly

In bioinformatics, *sequence assembly* refers to merging fragments of a much longer DNA sequence in order to reconstruct the original sequence.

Fragment DNA

Sequence the fragments and merge

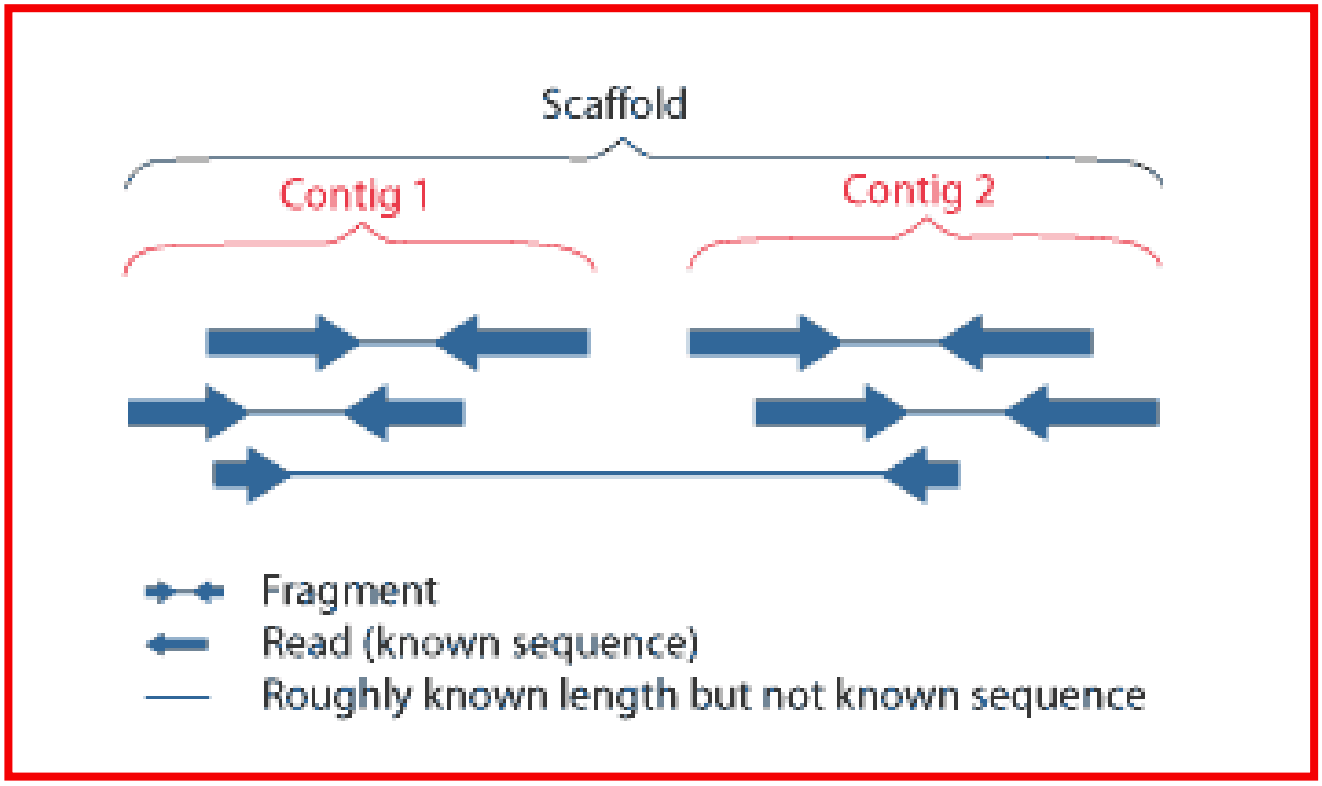


***De novo* short read assembly** is the process whereby we merge together individual sequence reads to form long contiguous sequences **'contigs'**, sharing the same nucleotide sequence as the original template DNA from which the sequence reads were derived.

Short read assembly requires to fragment the DNA molecule previous to NGS.

Contig vs scaffold

A contig (from contiguous) is a set of overlapping DNA fragments that together represent a consensus region of DNA.



A scaffold is composed of contigs and gaps.
Gap length can be guessed by incorporating information from paired ends or mate pairs of different insert sizes.

De novo short read assembly vs. short read mapping assembly (comparative assembly)

In sequence assembly, two different types can be distinguished:

1.- *de novo assembly*: assembling reads together so that they form a new, previously unknown sequence.

2.- *comparative assembly*: assembling reads against and existing backbone or reference sequence, building a sequence that is similar but not necessarily identical to the backbone sequence.

"De novo Assembly of a 40 Mb Eukaryotic Genome from Short Sequence Reads: Sordaria macrospora, a Model Organism for Fungal Morphogenesis"
http://www.plosgenetics.org/article/info%3Adoi%2F10.1371%2Fjournal.pgen.1000891

In tems of complexity and time requirements, *de novo* assemblers are orders of magnitude slower and more memory intensive than mapping assemblers. This is mostly due to the fact that the assembly algorithm need to compare every read with every other read.

N50

An N50 contig size of N means that **50% of the assembled bases are contained in contigs of length N or larger**.

N50 sizes are often used as a measure of assembly quality because they capture how much of the genome is covered by relatively large contigs.

An interesting *de novo* assembly study

RESEARCH ARTICLE OPEN ACCESS

De novo Assembly of a 40 Mb Eukaryotic Genome from Short Sequence Reads: *Sordaria macrospora*, a Model Organism for Fungal Morphogenesis

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Minou Nowrousian[1], Jason E. Stajich[2], Meiling Chu[3], Ines Engh[1], Eric Espagne[4], Karen Halliday[5], Jens Kamerewerd[1], Frank Kempken[6], Birgit Knab[7], Hsiao-Che Kuo[3¤], Heinz D. Osiewacz[7], Stefanie Pöggeler[8], Nick D. Read[3], Stephan Seiler[9], Kristina M. Smith[10], Denise Zickler[4], Ulrich Kück[1*], Michael Freitag[10*]

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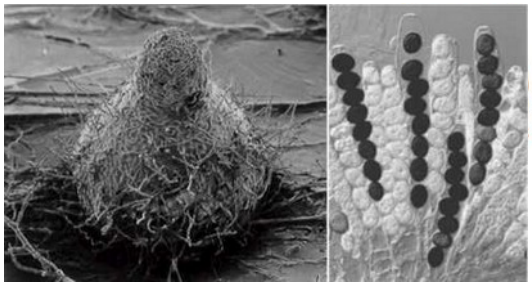
An interesting *de novo* assembly study



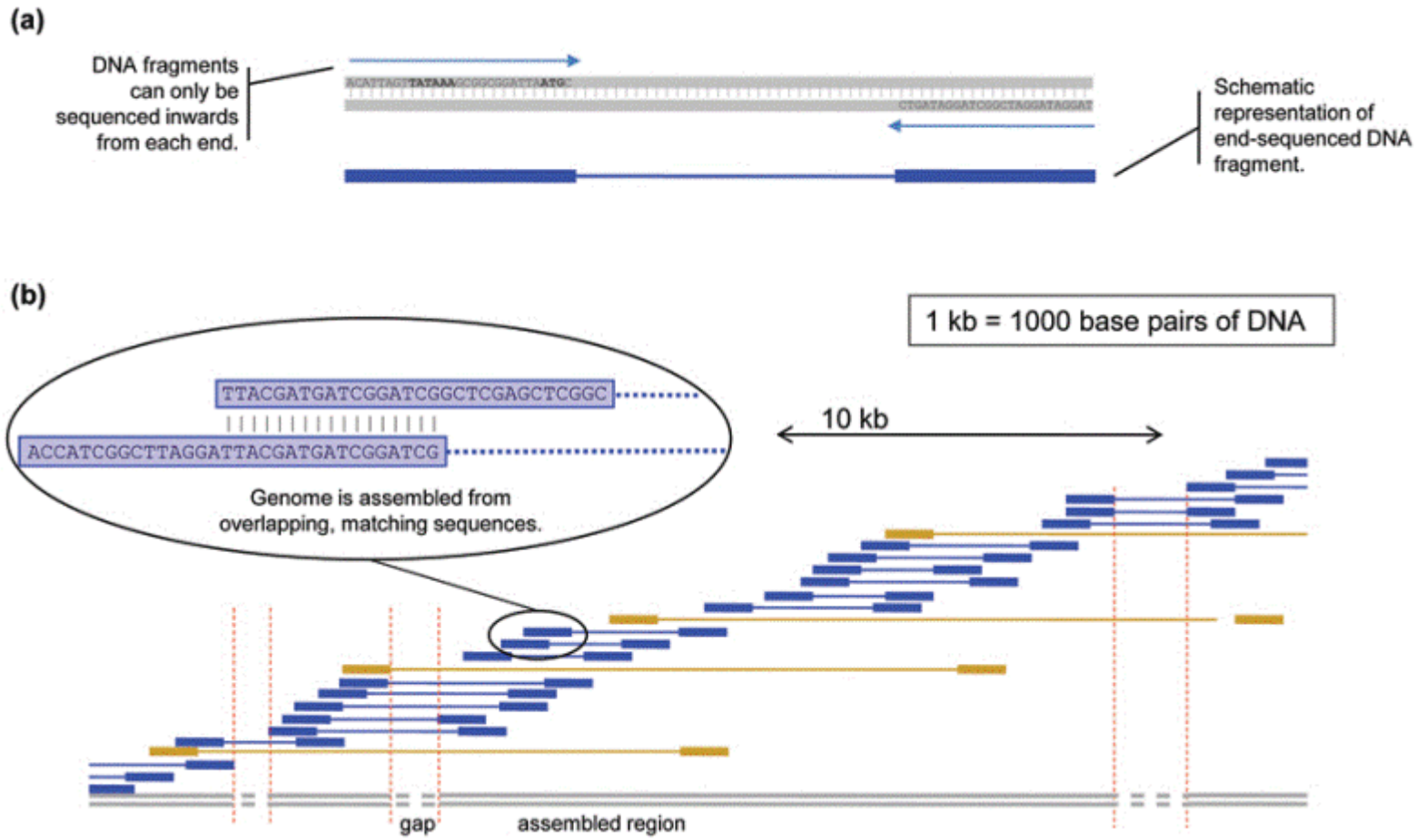
Filamentous fungi are of great importance in ecology, agriculture, medicine, and biotechnology. Thus, it is not surprising that genomes for more than 100 filamentous fungi have been sequenced, most of them by Sanger sequencing. While next-generation sequencing techniques have revolutionized genome resequencing, e.g. for strain comparisons, genetic mapping, or transcriptome and ChIP analyses, *de novo* assembly of eukaryotic genomes still presents significant hurdles, because of their large size and stretches of repetitive sequences. Filamentous fungi contain few repetitive regions in their 30–90 Mb genomes and thus are suitable candidates to test *de novo* genome assembly from short sequence reads. Here, we present a high-quality draft sequence of the *Sordaria macrospora* genome that was obtained by a combination of Illumina/Solexa and Roche/454 sequencing. Paired-end Solexa sequencing of genomic DNA to 85-fold coverage and an additional 10-fold coverage by single-end 454 sequencing resulted in ~4 Gb of DNA sequence. Reads were assembled to a 40 Mb draft version (N50 of 117 kb) with the Velvet assembler. Comparative analysis with *Neurospora* genomes increased the N50 to 498 kb. The *S. macrospora* genome contains even fewer repeat regions than its closest sequenced relative, *Neurospora crassa*. Comparison with genomes of other fungi showed that *S. macrospora*, a model organism for morphogenesis and meiosis, harbors duplications of several genes involved in self/nonself-recognition. Furthermore, *S. macrospora* contains more polyketide biosynthesis genes than *N. crassa*. Phylogenetic analyses suggest that some of these genes may have been acquired by horizontal gene transfer from a distantly related ascomycete group. Our study shows that, for typical filamentous fungi, *de novo* assembly of genomes from short sequence reads alone is feasible, that a mixture of Solexa and 454 sequencing substantially improves the assembly, and that the resulting data can be used for comparative studies to address basic questions of fungal biology.

An interesting *de novo* assembly study

Table 1. Main features of primary sequence data.

primary sequence data	Solexa	454	Solexa +454
no. of reads that were obtained	95,153,934	1,103,372	96,261,736
read length	36 nt	367 nt[1]	n.a.
total length of all sequence reads	3,426 Mb	415 Mb	3,879 Mb

1 average read length.



There are still gaps where the sequence is unknown, although the order of the sequenced sections relative to each other is known.

De novo short read assembly vs. short read mapping assembly

1)Coverage needs to increase to compensate for the decreased connectivity and produce a comparable assembly.

2)Certain problems cannot be overcome by deeper coverage: If a repetitive sequence is longer than a read, then coverage alone will never compensate, and all copies of that sequence will produce gaps in the assembly.

3)These gaps can be spanned by paired reads—consisting of two reads generated from a single fragment of DNA and separated by a known distance—as long as the pair separation distance is longer than the repeat.

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Assembly of large genomes using second-generation sequencing

Michael C. Schatz, Arthur L. Delcher and Steven L. Salzberg[1]

+ Author Affiliations

Abstract

Second-generation sequencing technology can now be used to sequence an entire human genome in a matter of days and at low cost. Sequence read lengths, initially very short, have rapidly increased since the technology first appeared, and we now are seeing a growing number of efforts to sequence large genomes de novo from these short reads. In this Perspective, we describe the issues associated with short-read assembly, the different types of data produced by second-gen sequencers, and the latest assembly algorithms designed for these data. We also review the genomes that have been assembled recently from short reads and make recommendations for sequencing strategies that will yield a high-quality assembly.

As genome sequencing technology has evolved, methods for assembling genomes have changed with it. Genome sequencers

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Article

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There is a Corrigendum (25 February 2010) associated with this document.

The sequence and *de novo* assembly of the giant panda genome

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Panda genome completed using short-read sequencing

By John Timmer | Last updated December 14, 2009 5:15 PM

Over the weekend, *Nature* released a paper that describes the genome of the giant panda. The results do tell us something about the animal itself—it's a vegetarian that looks like a carnivore—but is probably more notable for how it was obtained. It's the first genome to be completed using sequence from the relatively new Illumina platform, which typically reads less than 100 bases from each piece of DNA, but can read massive amounts of DNA in parallel.

Illumina isn't one of the technologies I've covered yet in our DNA sequencing series, but it's similar to some of the other ones, like Complete Genomics, in that the read length is very short (an average of 52 bases for the panda). But Illumina machines produce staggering amounts of these short reads in each run, and provided the authors with enough sequence to read each base, on average, 73 times. Figuring out how all these tiny pieces (all 3.3 billion of them) overlap is a serious computational problem, but the authors used a software package called SOAP that's specifically designed to handle these short reads.

The genome that resulted looks a lot like the one from the only other carnivore sequenced, the dog. Despite

37 paired-end sequence libraries, read length=52bp on average, average depth coverage per base =73

The sequence and *de novo* assembly of the giant panda genome

assembly of the giant panda genome will have far-reaching implications for promoting the construction of reference sequences for other animal and plant genomes in an efficient and cost-effective way.

Short-read *de novo* sequencing and assembly

For sequencing, we selected a 3-year-old female giant panda from the Chengdu breeding centre in China. The panda genome contains 20 pairs of autosomes and one pair of sex chromosomes ($2n = 42$) (Supplementary Fig. 1). We used a whole-genome shotgun sequencing strategy and Illumina Genome Analyser sequencing technology. DNA was extracted from the peripheral venous blood, and 37 paired-end sequencing libraries were constructed with insert sizes of about 150 base pairs (bp), 500 bp, 2 kb, 5 kb and 10 kb. In total, we generated 176-Gb of usable sequence (equal to 73-fold coverage of the whole genome), with an average read length of 52 bp (Supplementary Tables 1 and 2).

We assembled the short reads using SOAPdenovo (http://soap.genomics.org.cn)—a genome assembler developed specifically for use with next-generation short-read sequences[6] (Supplementary Fig. 2). SOAPdenovo uses the de Bruijn graph algorithm[7] and applies a stepwise strategy to make it feasible to assemble the panda genome using a supercomputer (32 cores and 512 Gb random access memory (RAM)). The algorithm is sensitive to sequencing errors, so we

The sequence and *de novo* assembly of the giant panda genome

Using next-generation sequencing technology alone, we have successfully generated and assembled a draft sequence of the giant panda genome. The assembled contigs (2.25 gigabases (Gb)) cover approximately 94% of the whole genome, and the remaining gaps (0.05 Gb) seem to contain carnivore-specific repeats and tandem repeats. Comparisons with the dog and human showed that the panda genome has a lower divergence rate. The assessment of panda genes potentially underlying some of its unique traits indicated that its bamboo diet might be more dependent on its gut microbiome than its own genetic composition. We also identified more than 2.7 million heterozygous single nucleotide polymorphisms in the diploid genome. Our data and analyses provide a foundation for promoting mammalian genetic research, and demonstrate the feasibility for using next-generation sequencing technologies for accurate, cost-effective and rapid *de novo* assembly of large eukaryotic genomes.

Top

De novo short read assembly

The genome that resulted looks a lot like the one from the only other carnivore sequenced, the dog. Despite some remaining gaps, the authors were able to find 99 percent of the dog genes in the panda genome, including enzymes typically involved in digesting meat. In contrast, they were unable to find any gene that looks like it would allow the panda to digest foliage, and they suggest we need to look to its gut bacteria to understand how it does that. So, although the panda is an exclusive herbivore behaviorally, its genome doesn't appear to have committed it to that lifestyle.

Available assemblers

Available assemblers

[edit

The following table lists assemblers that have a de-novo assembly capability on at least one of the supported technologies.[6]

Name	Type	Technologies	Author	Presented / Last updated	Licence*	Homepage
ABySS	(large) genomes	Solexa, SOLiD	Simpson, J. et al.	2008 / 2011	NC-A	link
ALLPATHS-LG	(large) genomes	Solexa, SOLiD	Gnerre, S. et al.	2011	OS	link
AMOS	genomes	Sanger, 454	Salzberg, S. et al.	2002? / 2008?	OS	link
Celera WGA Assembler / CABOG	(large) genomes	Sanger, 454, Solexa	Myers, G. et al.; Miller G. et al.	2004 / 2010	OS	link
CLC Genomics Workbench	genomes	Sanger, 454, Solexa, SOLiD	CLC bio	2008 / 2010	C	link
Cortex	genomes	Solexa, SOLiD	Iqbal, Z. *et al.*	2011	OS	link
DNA Dragon	genomes	Illumina, SOLiD, Complete Genomics, 454, Sanger	SequentiX	2011	C	link
DNAnexus	genomes	Illumina, SOLiD, Complete Genomics	DNAnexus	2011	C	link
Edena	genomes	Illumina	D. Hernandez, P. François, L. Farinelli, M. Osteras, and J. Schrenzel.	2008/2011	C	link
Euler	genomes	Sanger, 454 (,Solexa ?)	Pevzner, P. et al.	2001 / 2006?	(C / NC-A?)	link
Euler-sr	genomes	454, Solexa	Chaisson, MJ. et al.	2008	NC-A	link

Available assemblers

Forge	(large) genomes, EST, metagenomes	454, Solexa , SOLiD, Sanger	Platt, DM, Evers, D.	2010	OS	link
Geneious	genomes	Sanger, 454, Solexa	Biomatters Ltd	2009 / 2010	C	link
Graph Constructor	(large) genomes	Sanger, 454, Solexa, SOLiD	Convey Computer Corporation	2011	C	link
IDBA (Iterative De Bruijn graph short read Assembler)	(large) genomes	Sanger,454,Solexa	Yu Peng, Henry C. M. Leung, Siu-Ming Yiu, Francis Y. L. Chin	2010	(C / NC-A?)	link
MIRA (Mimicking Intelligent Read Assembly)	genomes, ESTs	Sanger, 454, Solexa	Chevreux, B.	1998 / 2011	OS	link
NextGENe	(small genomes?)	454, Solexa, SOLiD	Softgenetics	2008	C	link
Newbler	genomes, ESTs	454, Sanger	454/Roche	2009	C	link
PASHA	(large) genomes	Illumina	Liu, Schmidt, Maskell	2011	OS	link
Phrap	genomes	Sanger, 454, Solexa	Green, P.	1994 / 2008	C / NC-A	link
TIGR Assembler	genomic	Sanger	-	1995 / 2003	OS	link
Ray[7]	genomes	Illumina, mix of Illumina and 454, paired or not	Sébastien Boisvert, François Laviolette & Jacques Corbeil.	2010	OS [GNU General Public License]	link
Sequencher	genomes	traditional and next generation sequence data	Gene Codes Corporation	1991 / 2009 / 2011	C	link
SeqMan NGen	(large) genomes, exomes, transcriptomes, metagenomes, ESTs	Illumina, ABI SOLiD, Roche 454, Ion Torrent, Solexa, Sanger	DNASTAR	2007 / 2011	C	link
SHARCGS	(small) genomes	Solexa	Dohm et al.	2007 / 2007	OS	link
SOPRA	genomes	Illumina, SOLiD, Sanger, 454	Dayarian, A. et al.	2010 / 2011	OS	link

Available assemblers

SSAKE	(small) genomes	Solexa (SOLiD? Helicos?)	Warren, R. et al.	2007 / 2007	OS	link
SOAPdenovo	genomes	Solexa	Li, R. et al.	2009 / 2009	OS	link
Staden gap4 package	BACs (, small genomes?)	Sanger	Staden et al.	1991 / 2008	OS	link
Taipan	(small) genomes	Illumina	Schmidt, B. *et al.*	2009	OS	link
VCAKE	(small) genomes	Solexa (SOLiD?, Helicos?)	Jeck, W. et al.	2007 / 2007	OS	link
Phusion assembler	(large) genomes	Sanger	Mullikin JC, *et al.*	2003	OS	link
Quality Value Guided SRA (QSRA)	genomes	Sanger, Solexa	Bryant DW, *et al.*	2009	OS	link
Velvet	(small) genomes	Sanger, 454, Solexa, SOLiD	Zerbino, D. et al.	2007 / 2009	OS	link

*__Licences:__ OS = Open Source; C = Commercial; C / NC-A = Commercial but free for non-commercial and academics; Brackets = unclear, but most likely C / NC-A

Genomic DNA assembly vs ESTs assembly

ESTs

An *expressed sequence tag* or EST is a short sub-sequence of a cDNA sequence.

Because these clones consist of DNA that is complementary to mRNA, the ESTs represent portions of expressed genes.

Many distinct ESTs are often partial sequences that correspond to the **same** mRNA of an organism.

source: Wikipedia

Genomic DNA assembly vs ESTs assembly

Typically, the short fragments, reads, result from shotgun sequencing of *genomic DNA or gene transcripts (ESTs)*.

To deal with these two problems, there are Genome assemblers and EST assemblers.

EST assemblers differ from genome assemblers in several ways**. *The sequence for EST assembly are the transcribed mRNA of a cell and represent only a subset of the whole genome*. ESTs do not usually contain repeats**, **since they represent gene transcripts**, **and repeats are mainly located in intergenic regions.**

Parallel problems for EST assembly:

1.- Cells tend to have a certain number of genes that are constantly expressed in very high amounts (housekeeping genes), which leads to the problem of similar sequences present in high amounts in the data set to be assembled.

2.- Genes sometimes overlap in the genome (sense-antisense transcription), and should ideally still be assembled separately.

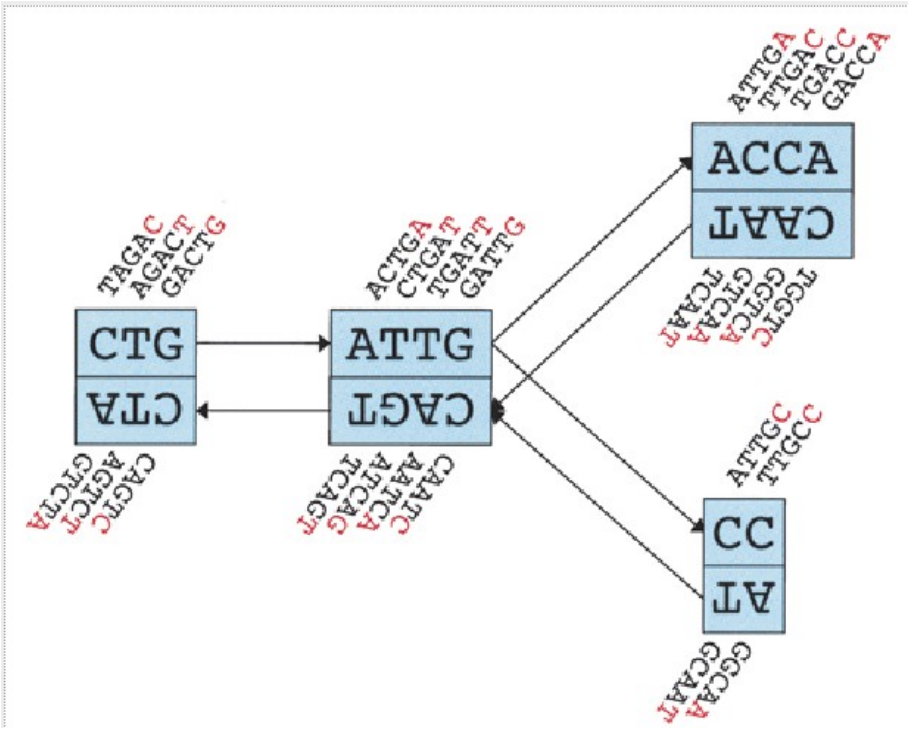
3.- EST assembly is also complicated by features like (cis-) alternative splicing, trans-splicing, SNPs and post-transcriptional modification.

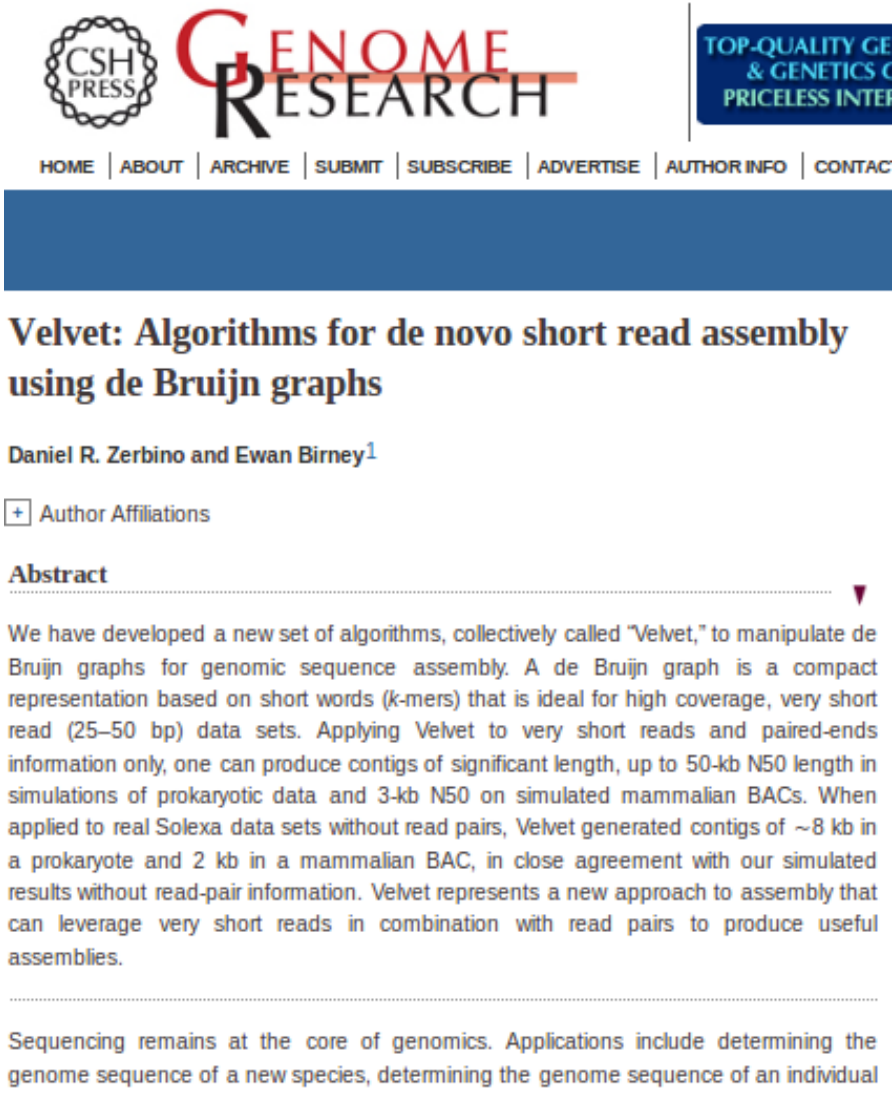
*** Housekeeping gene - typically a constitutive gene that is transcribed at a relatively constant level across many or all known conditions. The housekeeping gene's products are typically needed for maintenance of the cell. It is generally assumed that their expression is unaffected by experimental conditions. Examples include actin, GAPDH and ubiquitin.

Sequence Mapping and Assembly Assessment Project (SMAAP)

Initiative to compare and evaluate the best tools for mapping and assembly.

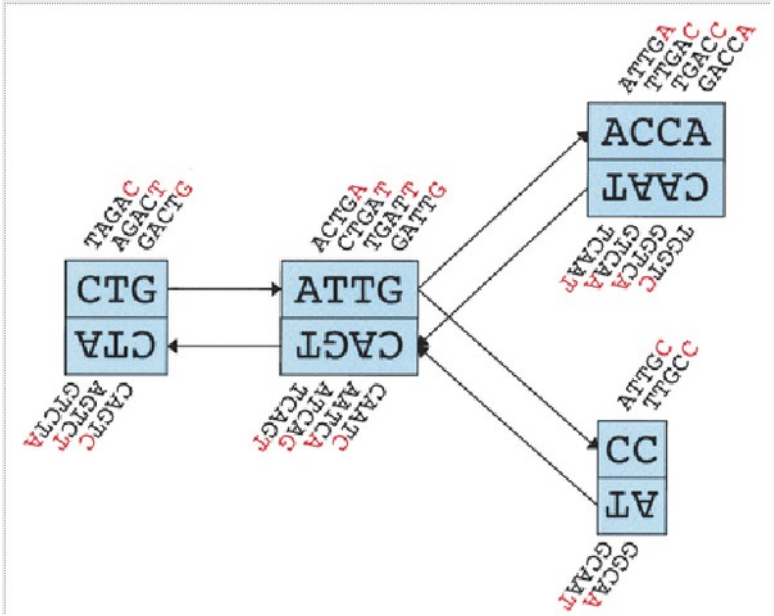
Velvet: Using de Bruijn graphs for *de novo* short read assembly





***Velvet needs about 20-25x coverage and paired reads

Velvet: Using de Bruijn graphs for *de novo* short read assembly



In this representation of data, elements are not organized around reads, but around words of k nucleotides, or k-mers.
(k-mer length = hash length = length in base pairs of the words being hashed)

Reads are mapped as paths through the graph, going from one word to the next in a determined order.

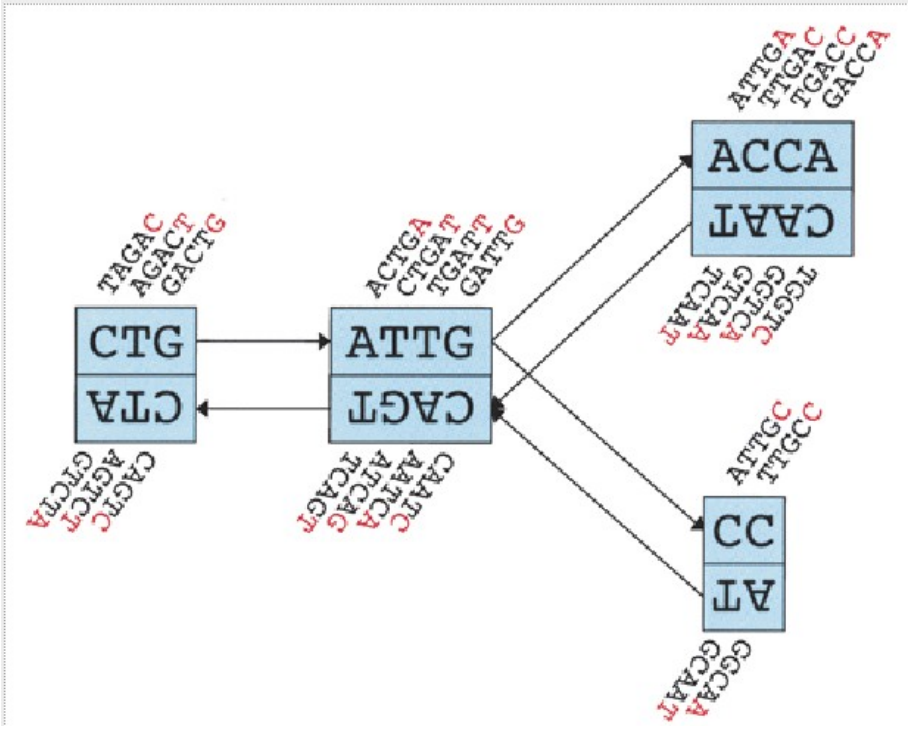
The fundamental data structure in the de Bruijn graph is based on k-mers, not reads, thus high redundancy is naturally handled by the graph without affecting the number of nodes.

<u>In the de Bruijn graph, each node N represents a series of overlapping k-mers. Adjacent k-mers overlap by k − 1 nucleotides.</u> The marginal information contained by a k-mer is its last nucleotide. The sequence of those final nucleotides is called *the sequence of the node*, or s(N).

Each node N is attached to a twin node N, which represents the reverse series of reverse complement k-mers. This ensures that overlaps between reads from opposite strands are taken into account. Note that the sequences attached to a node and its twin do not need to be reverse complements of each other.
The union of a node N and its twin N is called a "block." Any change to a node is implicitly applied symmetrically to its twin. A block therefore has two distinguishable sides.

Velvet: Using de Bruijn graphs for *de novo* short read assembly

Nodes can be connected by a directed "arc." In that case, *the last k-mer of an arc's origin node overlaps with the first of its destination node*. Because of the symmetry of the blocks, if an arc goes from node A to B, a symmetric arc goes from Graphic to Graphic. Any modification of one arc is implicitly applied symmetrically to its paired arc.



Schematic representation of our implementation of the de Bruijn graph. Each node, represented by a single rectangle, represents a series of overlapping *k*-mers (in this case, *k* = 5), listed directly *above* or *below*. (Red) The last nucleotide of each *k*-mer. The sequence of those final nucleotides, copied in large letters in the rectangle, is the sequence of the node. The twin node, directly attached to the node, either *below* or *above*, represents the reverse series of reverse complement *k*-mers. Arcs are represented as arrows *between* nodes. The last *k*-mer of an arc's origin overlaps with the first of its destination. Each arc has a symmetric arc. Note that the two nodes on the *left* could be merged into one without loss of information, because they form a chain.

Exercise: perform a *de novo* assembly with a set of sequences from *Pseudomonas*

Iniciamos Docker con las variables necesarias para enlazar los directorios/carpetas con los datos de input (samples) y los datos de output (resultados), como se muestra a continuación:

```
SAMPLES_LOCAL=/Users/ograna/BIOINFORMATICS/Curso_UFV/De-Novo-Genome-Assembly/Exercises_for_students/
SAMPLES_DOCKER=/SAMPLES
RESULTS_LOCAL=/Users/ograna/BIOINFORMATICS/Curso_UFV/De-Novo-Genome-Assembly/Exercises_for_students/RESULTS
RESULTS_DOCKER=/RESULTS
```

Arrancamos Docker exportando el display gráfico

```
xhost +
docker run --rm  -e DISPLAY=$DISPLAY -v /tmp/.X11-unix/:/tmp/.X11-unix/  -v $SAMPLES_LOCAL:$SAMPLES_DOCKER -v $RESULTS_LOCAL:$RESULTS_DOCKER -it osvaldogc/ufv:2.0 /bin/bash
```

Exercise: perform a *de novo* assembly with a set of sequences from *Pseudomonas*

reads file : pseudomonas.fq (36bp reads, paired-end)

****how many pairs of paired-end reads are contained in the file?

grep -c '^@' pseudomonas.fq

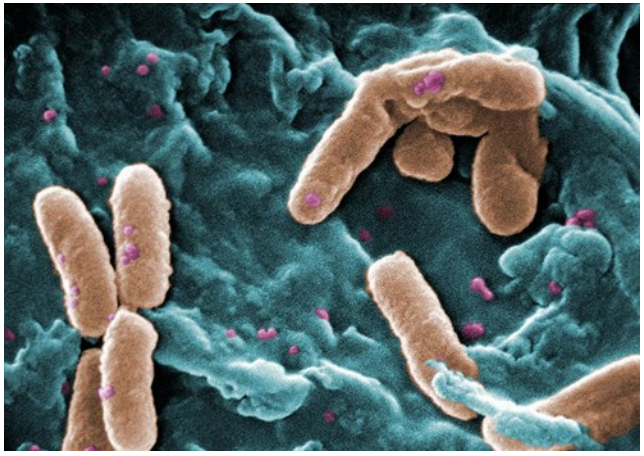
1.- Build the hash table for the reads with velvet (velveth)

velveth ENSAMBLAJE 21 -shortPaired -fastq pseudomonas.fq

ENSAMBLAJE: directory name for the output files
21: K-mer length or hash length
pseudomonas.fq -> paired-end reads in fastq format

2.- Build the graph with velvet (velvetg)

velvetg ENSAMBLAJE -unused_reads yes



Exercise: perform a *de novo* assembly with a set of sequences from *Pseudomonas*

How many contigs do we get?

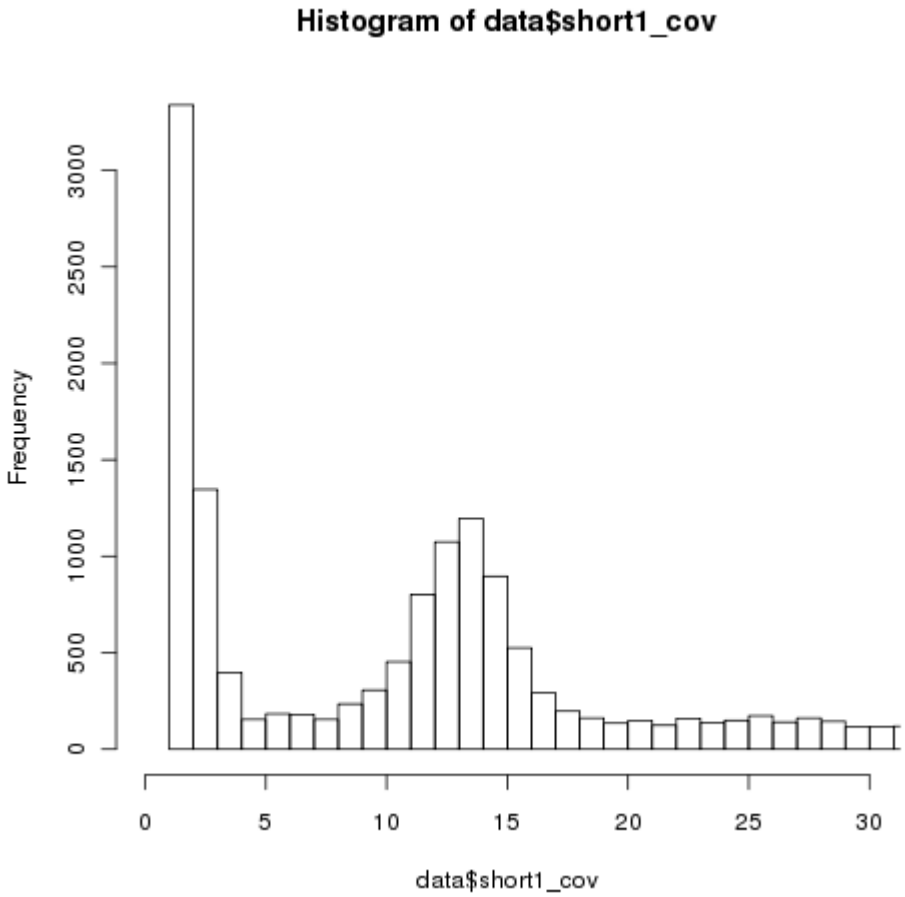
Exercise: perform a *de novo* assembly with a set of sequences from *Pseudomonas*

3.- From the ENSAMBLAJE directory, execute R:

```
cd ENSAMBLAJE

R
> data=read.table("stats.txt",header=TRUE)
> hist(data$short1_cov,xlim=range(0,30),breaks=5e5)
```

what we see in the plot is the frecuency of contigs (Y axis) with a specific k-mer coverage (X axis)



Exercise: perform a *de novo* assembly with a set of sequences from *Pseudomonas*

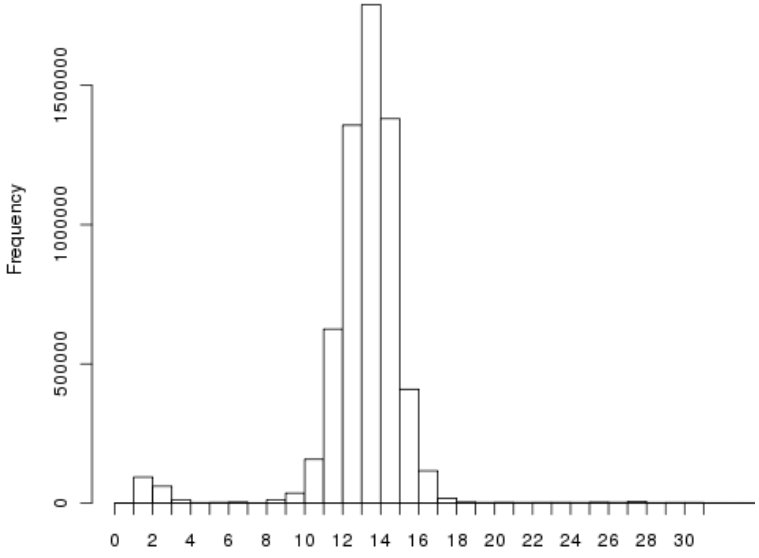
4.- From the ENSAMBLAJE directory, execute R:

```
R
> library(plotrix)
> data=read.table("stats.txt",header=TRUE)
> weighted.hist(data$short1_cov,data$lgth,breaks=0:100,xlim=range(0,30))
```

***to install this module from R: install.packages("plotrix")

in this plot we have weighted the coverage with the node lengths. Below 7x or 8x we find mainly short and low coverage nodes, which are likely to be errors.

From the weighted histogram it must be pretty clear that the expected coverage of contigs is near 14x.



Exercise: perform a *de novo* assembly with a set of sequences from *Pseudomonas*

5.- Rebuilding the graph with the expected coverage:

velvetg ENSAMBLAJE -exp_cov 14 -cov_cutoff 7 -amos_file yes

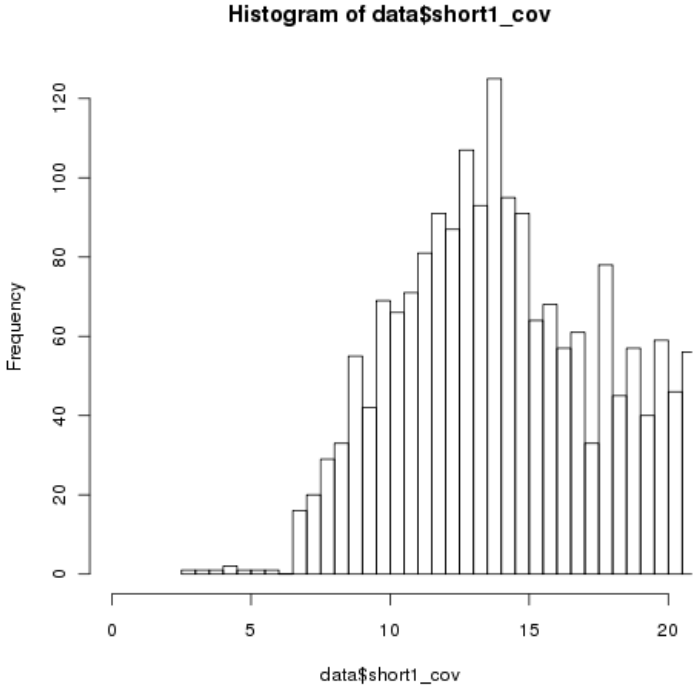
How many contigs do we get now?

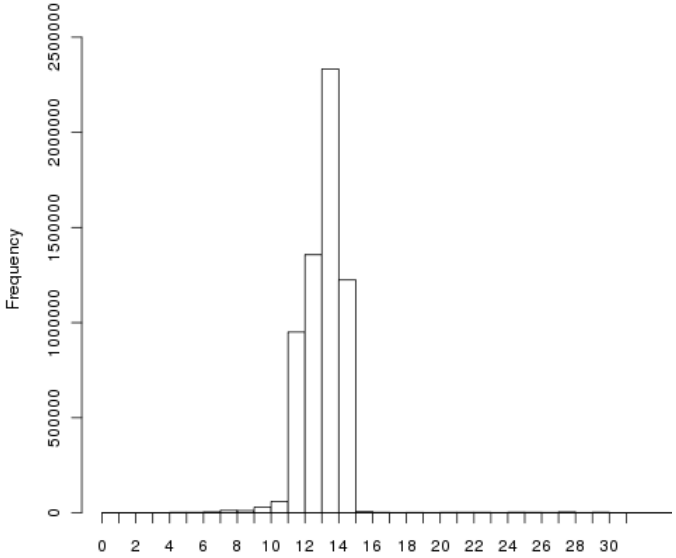
Exercise: perform a *de novo* assembly with a set of sequences from *Pseudomonas*

5.- From the test directory, execute R:

```
R
> library(plotrix)
> data=read.table("stats.txt",header=TRUE)
> hist(data$short1_cov,xlim=range(0,20),breaks=1000000)
> weighted.hist(data$short1_cov,data$lgth,breaks=0:100,xlim=range(0,30))
```

now the obtained contigs are much bigger than before.





Exercise: perform a *de novo* assembly with a set of sequences from *Pseudomonas*

We might want to save the graph generated with R:

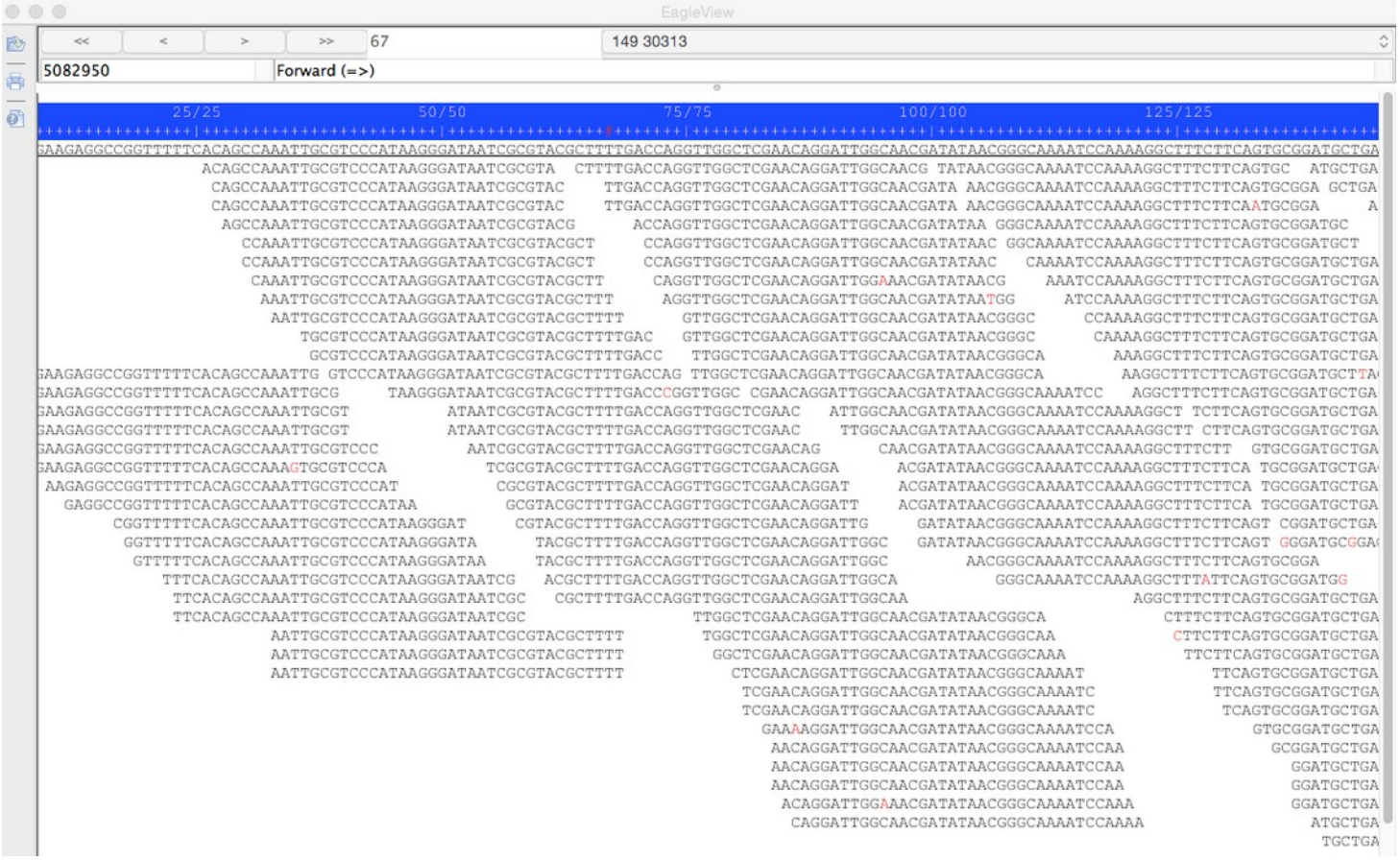
```
> png(file="myGraph.png")
> hist(data$short1_cov,xlim=range(0,30),breaks=5e5)
> dev.off()
> q()
```

How to view assembled contigs with EagleView

1)Convert the velvet afg output file to ace format
(requires to have the AMOS package installed)

perl amos2ace.pl velvet_asm.afg -o velvet_asm.ace

2) Open the ace file
with EagleView



Recommended references

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* Li R, Zhu H, Ruan J, Qian W, Fang X, Shi Z, Li Y, Li S, Shan G, Kristiansen K, Li S, Yang H, Wang J, Wang J. *De novo assembly of human genomes with massively parallel short read sequencing*. Genome Res. 2010 Feb;20(2):265-72.

* Li R, Fan W, Tian G, Zhu H, He L, Cai J, Huang Q, Cai Q, Li B, Bai Y, Zhang Z, Zhang Y, Wang W, Li J, Wei F, Li H, Jian M, Li J, Zhang Z, Nielsen R, Li D, Gu W, Yang Z, Xuan Z, Ryder OA, Leung FC, Zhou Y, Cao J, Sun X, Fu Y, Fang X, Guo X, Wang B, Hou R, Shen F, Mu B, Ni P, Lin R, Qian W, Wang G, Yu C, Nie W, Wang J, Wu Z, Liang H, Min J, Wu Q, Cheng S, Ruan J, Wang M, Shi Z, Wen M, Liu B, Ren X, Zheng H, Dong D, Cook K, Shan G, Zhang H, Kosiol C, Xie X, Lu Z, Zheng H, Li Y, Steiner CC, Lam TT, Lin S, Zhang Q, Li G, Tian J, Gong T, Liu H, Zhang D, Fang L, Ye C, Zhang J, Hu W, Xu A, Ren Y, Zhang G, Bruford MW, Li Q, Ma L, Guo Y, An N, Hu Y, Zheng Y, Shi Y, Li Z, Liu Q, Chen Y, Zhao J, Qu N, Zhao S, Tian F, Wang X, Wang H, Xu L, Liu X, Vinar T, Wang Y, Lam TW, Yiu SM, Liu S, Zhang H, Li D, Huang Y, Wang X, Yang G, Jiang Z, Wang J, Qin N, Li L, Li J, Bolund L, Kristiansen K, Wong GK, Olson M, Zhang X, Li S, Yang H, Wang J, Wang J. *The sequence and de novo assembly of the giant panda genome*. Nature. 2010 Jan 21;463(7279):311-7. Epub 2009 Dec 13. Erratum in: Nature. 2010 Feb 25;463(7284):1106.